

Lucas A. Saavedra

COMPUTER ENGINEER · AI RESEARCHER

Buenos Aires, Argentina

☎ (+54) 11-5574-2466 | ✉ lucasarielsaavedra@hotmail.com | 🏠 lucassaavedra123.github.io | 📱 lucassaavedra123 | 🌐 lucassaave

Experience

Accenture

Buenos Aires, Argentina

SENIOR FULL-STACK AI DEVELOPER

May 2025 - PRESENT

- GenAI Studio team: LLM-based multi-agent systems (LangGraph, FastAPI, Azure OpenAI, Elasticsearch) for automated code migration between languages, data-quality testing, and AI-assisted accounting reconciliation.
- Social media analytics platform: Neo4j knowledge graph, Spring Boot batch classification pipeline, and a LangGraph conversational agent exposed through a Next.js chat interface, including an MCP server to expose the graph as tools for the agent.
- Real-time computer vision systems (YOLO, OpenCV, TensorFlow) for vehicle detection, customer profiling, and dataset preparation for gas station clients.
- Full-stack development with Next.js/React and enterprise Angular applications, deployed with Docker and Azure Pipelines.
- **Technologies used: Python, Large Language Models (LLMs), LangGraph, FastAPI, Azure OpenAI, Elasticsearch, Neo4j, Spring Boot, Next.js, React, Angular, Docker, Azure Pipelines, Computer Vision (YOLO, OpenCV, TensorFlow).**

BIOMED UCA-CONICET, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

RESEARCH ASSISTANT

Jan. 2022 - PRESENT

- Implemented Deep Learning models for static and dynamic analysis of the nicotinic acetylcholine receptor (nAChR), with orientation to graph theory, recurrent networks and geometric computing.
- Analyzed distribution and density of nicotinic acetylcholine receptors in the plasma membrane using super-resolution microscopy (STED, STORM), with advanced analytics implemented in MATLAB based on graph theory and computational geometry.
- Developed an efficient and autonomous algorithm based on supervised Graph Neural Networks (GNNs) in TensorFlow to detect and quantify dynamic particle clustering in single-molecule microscopy (SMLM) data.
- Applied AlphaFold3-equivalent models to study protein-ligand structures (nAChR and cholesterol), combined with molecular dynamics (MD) simulations using GROMACS on high-performance computing (HPC) platforms with SLURM.
- **Technologies used: Python, Java, TensorFlow, PyTorch, Pandas, NumPy, Docker, MongoDB, MATLAB, GROMACS, SLURM.**

Data Science and AI Laboratory, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

RESEARCH ASSISTANT

Jan. 2025 - June 2025

- Supervisor: Prof. Antonio Quintero-Rincón.
- Analysis of electroencephalogram (EEG) data during mental arithmetic tasks using self-supervised learning—specifically, hierarchical variational autoencoders—to detect specific cognitive activity with PyTorch.

Facultad de Ingeniería, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

ASSISTANT PROFESSOR

Mar. 2023 - June 2025

- "Introduction to Data Science": Artificial Intelligence concepts, Neural Networks, Scientific Method.
- "Data-Oriented Programming": Object-Oriented Programming, Python, Java, NumPy, Pandas, UML Software Design.
- "Discrete Mathematics": Number Theory, Graph Theory, Proof Methods, Boolean Algebra.

UpWork

FREELANCER

Dec. 2023 - May 2025

- Fixed and hourly contracts.
- Web scraping with Python and Selenium in AWS EC2 Linux instances.
- YOLO training to identify basketball players in video matches from a European basketball league, with a SQL database hosted in Supabase.
- GUIs developed with built-in Napari plugins to gather information from laser spot images, combined with decision tree classifiers in scikit-learn.
- **Technologies used: Python, YOLO, Pandas, Docker, SQL, EC2 AWS, Selenium, Napari, scikit-learn.**

Deloitte

Buenos Aires, Argentina

BACK-END DEVELOPER

Jun. 2021 - Jun. 2022

- Developed Codex Gigas, a malware "DNA" search engine that identifies patterns and characteristics for malware hunting using REST APIs with Python.
- I worked in Agile environments.
- Built ImageSimilarity, a tool to classify images based on similarity using the metric SSIM.
- Developed applications for ThreatConnect using Python, Docker, MongoDB, and SQL.
- **Technologies used: Python, Docker, MongoDB, SQL, Redis.**

Deloitte

Buenos Aires, Argentina

CYBER RISK ADVISORY INTERN

Jan. 2021 - Jun. 2021

- Offensive/Defensive Security and Threat Intelligence basic concepts.

Skills

DevOps AWS, Docker, CircleCI
Technologies Django, REST API, TensorFlow, PyTorch, Pandas, NumPy, SQL, MongoDB, Redis, Selenium, Docker, YOLO, Langgraph, Langchain, Azure OpenAI
Programming Python, JAVA, C, C++, R, LaTeX, MATLAB

Education

Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

COMPUTER ENGINEERING

2018 - 2023

- **Final work:** Comparison of Machine Learning models for the analysis of anomalous diffusion of acetylcholine neurotransmitter receptor trajectories. (Grade: 10)
- **GPA:** 8.02/10.

Publications and conferences

- 2022 **Mapping the nicotinic acetylcholine receptor nanocluster topography at the cell membrane with sted and storm nanoscopies**, International Journal of Molecular Sciences
- 2023 **The individual molecule and its tribe, the nanocluster: The case of the nicotinic acetylcholine receptor**, 11th International Weber Symposium *Punta del Este, Uruguay*
- 2024 **A supervised graph-based deep learning algorithm to detect and quantify clustered particles**, Nanoscale
- 2024 **Machine learning empowers static and dynamics clustering characterization of transmembrane receptors**, Building Bridges in Computational Biophysics (BBCB) v3.0
- 2025 **Temporal convolutional networks work as general feature extractors for single-particle diffusion analysis**, Journal of Physics: Photonics
- 2025 **Concurrent diffusion of nicotinic acetylcholine receptors and fluorescent cholesterol disclosed by two-colour sub-millisecond minflux-based single-molecule tracking**, Nature Communications
- 2025 **Quantitative evaluation of methods to analyze motion changes in single-particle experiments**, Nature Communications
- 2025 **Environmental enrichment delays the onset of cognitive decline in a transgenic mouse model of Alzheimer disease**, XL Annual Meeting SAN
- 2026 **Machine Learning in Single-Molecule Tracking Analysis of Superresolution Optical Microscopy Data**, Cells

Certifications and Courses

- 2020 **The Data Scientist's Toolbox**, Johns Hopkins University, Coursera
- 2020 **Neural Networks and Deep Learning**, DeepLearning.AI, Coursera
- 2021 **Machine Learning**, Stanford Online, Coursera
- 2021 **Programming Foundations: Design Patterns (2013)**, Linkedin Learning
- 2021 **R Programming**, Johns Hopkins University, Coursera
- 2022 **Natural Language Processing in TensorFlow**, DeepLearning.AI, Coursera
- 2022 **DeepLearning.AI Tensorflow Developer**, DeepLearning.AI, Coursera
- 2022 **AWS Certified Cloud Practitioner**,
- 2024 **Deep Neural Networks with PyTorch**, IBM, Coursera
- 2025 **Introduction to Statistics**, Stanford Online, Coursera
- 2025 **Duolingo English Test (DET) Score: 135 (Advanced: CEFR C1)**,
- 2025 **Bioinformatics; Learn Docking & Mol Dynamics Simulation**, Udemy
- 2025 **Basic to Advanced: Retrieval-Augmented Generation (RAG)**, Udemy
- 2025 **IELTS Score: 7.5**,